

FINM 25000 Homework 2: Technical Indicators & Strategy Backtesting

Ticker: AAPL

Period: 5 years of daily OHLCV bars

Data source: Alpaca historical market data

Initial capital: \$100,000

Constraints: long-only, no leverage, no short selling

Strategy Descriptions

Buy & Hold: fully invested benchmark used as the baseline comparison.

Trend Following: buys when MACD is above its signal line, ADX is above 25, and price is above the 50-day SMA. It exits when MACD crosses below its signal or price loses the 50-day SMA.

Mean Reversion: buys oversold conditions where RSI is below 30 and price is below the lower Bollinger Band. It exits when RSI is above 70 or price moves above the upper Bollinger Band.

Custom Strategy: combines trend, momentum, volatility, and volume. It requires $EMA_{20} > SMA_{50}$, $MACD > signal$, RSI between 45 and 70, price above the Bollinger midline, positive CMF, and a normal ATR regime. It exits on trend, momentum, or volume deterioration.

Performance Comparison

Strategy	Total Return	CAGR	Volatility	Sharpe	Sortino	Max DD
Win Rate						
Buy & Hold	106.3%	15.6%	27.6%	0.67	0.97	-33.3%
100.0%						
Trend Following	2.8%	0.6%	9.2%	0.11	0.07	-13.0%
41.4%						
Mean Reversion	15.1%	2.9%	13.9%	0.27	0.16	-25.2%
100.0%						
Custom	42.0%	7.3%	10.2%	0.74	0.60	-11.2%
40.7%						

Discussion

The highest Sharpe ratio in this run belongs to Custom. The key comparison is risk-adjusted performance, not just raw return, so the equity curve and drawdown chart should be reviewed together with the Sharpe, Sortino, and maximum drawdown metrics.

Charts

The repository includes HTML chart exports in the charts/ folder:

- price chart with indicators and buy/sell signals
- equity curve comparison
- drawdown comparison